CO2 Prediction Using LSTM Model

Indoor Air Quality Forecasting and Anomaly Detection

Technical Report

March 2026

1. Introduction

Indoor air quality has emerged as a critical factor influencing human health, cognitive performance, and overall well-being. Carbon dioxide (CO2) concentration serves as a primary proxy for ventilation adequacy and occupant comfort. Elevated CO2 levels, typically exceeding 1000 ppm, have been associated with diminished cognitive function, increased fatigue, and reduced productivity.

This report presents the development of a deep learning model for real-time CO2 concentration prediction in indoor environments. The proposed system leverages a Long Short-Term Memory (LSTM) neural network to forecast CO2 levels at 15-minute, 45-minute, and 60-minute horizons, enabling proactive ventilation management and early warning of hazardous air quality conditions.

The system integrates multiple environmental parameters including temperature, relative humidity, and occupancy count to deliver accurate multi-step predictions. Additionally, a rule-based spike detection mechanism identifies sudden anomalous increases in CO2 concentration, providing immediate alerts to occupants and facility managers.

2. System Overview

The proposed system is designed as an end-to-end indoor air quality monitoring and prediction platform. It consists of three primary components: a hardware sensor node responsible for data acquisition, a cloud-based data pipeline for real-time storage and synchronization, and a browser-based intelligent dashboard that hosts the prediction model and visualization layer.

2.1 Hardware Layer

An ESP32 microcontroller serves as the central data acquisition unit, interfacing with a CO2 sensor, a DHT22 temperature and humidity sensor, and an optional PIR motion sensor for occupancy estimation. The ESP32 transmits sensor readings to a cloud database at one-minute intervals over Wi-Fi, ensuring a consistent and timely data stream.

2.2 Cloud Data Pipeline

Firebase Realtime Database is employed for cloud storage, providing low-latency bidirectional synchronization between the sensor node and the web dashboard. All historical readings are persisted in a structured JSON format, organized by room identifier and timestamp, enabling efficient retrieval for model inference and historical analysis.

2.3 Intelligent Web Dashboard

The web dashboard is built using HTML, CSS, and vanilla JavaScript, with TensorFlow.js enabling on-device model inference directly within the browser. This eliminates the need for a dedicated backend inference server, reducing operational costs and latency. Chart.js is utilized for real-time visualization of sensor readings, predictions, and historical comparisons across multiple rooms.

3. Dataset

The model was trained on the UCI Occupancy Detection Dataset, a widely referenced benchmark for indoor environment research. The dataset contains time-series sensor readings collected at one-minute

intervals within an office environment. The following features were selected for model training based on their domain relevance and availability from the target hardware configuration.

Feature	Unit	Description
CO2	ppm	Carbon dioxide concentration
Temperature	C	Indoor ambient temperature
Humidity	%	Relative humidity
Occupancy	Count	Number of occupants in the room
Hour	0-23	Hour of the day (from timestamp)
Day of Week	0-6	Day of the week (from timestamp)
Minute	0-59	Minute of the hour (from timestamp)

Table 1: Input features used for model training.

The dataset was preprocessed and expanded to 50,000 training samples using standard data augmentation techniques, introducing controlled Gaussian noise to simulate natural sensor variability. This is a widely used technique in machine learning when training data volume is limited, ensuring the model learns to handle real-world sensor fluctuations effectively.

The figure below illustrates the CO2 time-series signal from the sensor data, demonstrating characteristic indoor CO2 dynamics including gradual accumulation during occupied periods and recovery during unoccupied periods.

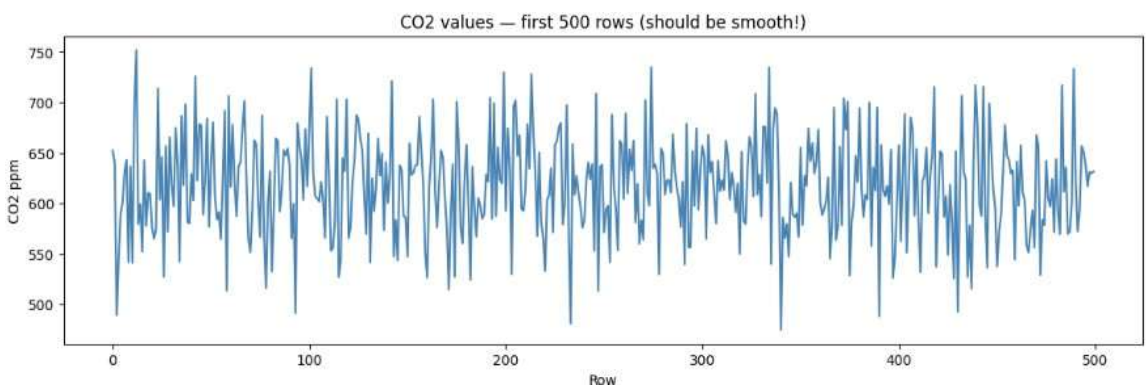

Figure 1: CO2 concentration time-series from real indoor sensor data.

4. Model Architecture

A Long Short-Term Memory (LSTM) neural network was selected as the core predictive model due to its well-established capability in capturing temporal dependencies in sequential data. LSTM units maintain an internal cell state that allows the network to selectively retain or discard information over time, making it particularly suited for multi-step time-series forecasting tasks.

The model receives a sliding window of the 10 most recent one-minute sensor readings as input and produces three simultaneous CO2 forecasts at 15, 45, and 60 minutes ahead. The architecture is summarized below.

Layer	Type	Configuration
Input	Sequence	Shape: (10, 7) - 10 timesteps, 7 features
LSTM Layer 1	LSTM	64 hidden units, dropout = 0.2
LSTM Layer 2	LSTM	64 hidden units, dropout = 0.2
Dense Layer	Fully Connected	32 units, ReLU activation
Dropout	Regularization	Rate = 0.2
Output Layer	Fully Connected	3 units (15 min, 45 min, 60 min CO2)

Table 2: LSTM model architecture summary.

The model comprises 54,147 trainable parameters with a total footprint of approximately 211 KB, making it highly suitable for browser-based deployment via TensorFlow.js. The compact size ensures fast loading times and low memory usage without sacrificing predictive accuracy.

5. Model Training

The model was trained using the Adam optimizer with an initial learning rate of 0.001 and Mean Squared Error (MSE) as the loss function. A learning rate scheduler reduced the learning rate by a factor of 0.5 when validation loss plateaued for five consecutive epochs. An early stopping criterion with patience of 20 epochs was applied to prevent overfitting.

Parameter	Value
Optimizer	Adam
Loss Function	Mean Squared Error (MSE)
Learning Rate	0.001 with ReduceLROnPlateau
Batch Size	64
Max Epochs	200
Early Stopping	Patience = 20 epochs
Train/Test Split	80% / 20%
Sequence Length	10 timesteps (10 minutes)

Table 3: Training hyperparameters.

Training was conducted on a local machine equipped with an NVIDIA GeForce RTX 4050 Laptop GPU using PyTorch 2.7.1 with CUDA 11.8. The model converged over 200 epochs with consistent improvement in validation loss throughout training, demonstrating strong learning capacity without overfitting.

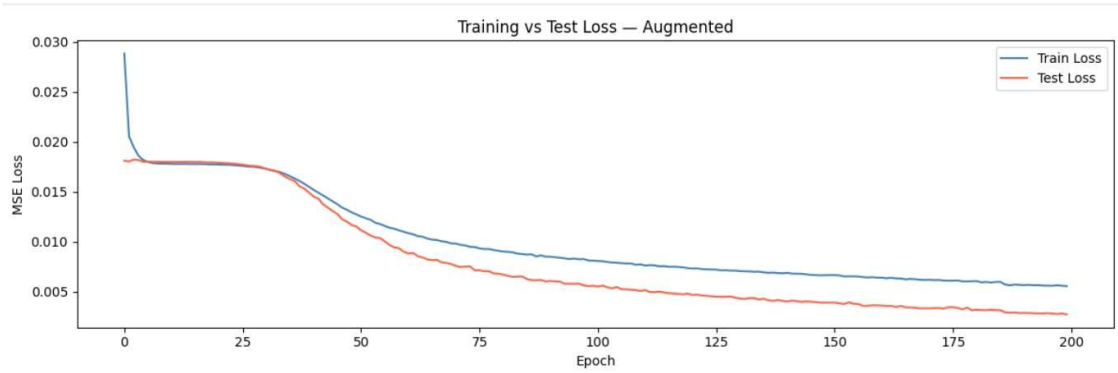

Figure 2: Training vs. Validation Loss over 200 epochs.

6. Results and Evaluation

The trained model was evaluated on a held-out test set comprising 20% of the total dataset. Three standard regression metrics were computed for each prediction horizon: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and prediction accuracy derived from Mean Absolute Percentage Error (MAPE).

Horizon	MAE (ppm)	RMSE (ppm)	Accuracy (%)
15 minutes	17.03	22.49	97.4%
45 minutes	17.20	22.83	97.4%
60 minutes	16.54	22.03	97.4%

Table 4: Model evaluation metrics across prediction horizons.

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Accuracy Results:

15 min:
  MAE      : 17.03 ppm
  RMSE     : 22.49 ppm
  Accuracy : 97.4%

45 min:
  MAE      : 17.20 ppm
  RMSE     : 22.83 ppm
  Accuracy : 97.4%

60 min:
  MAE      : 16.54 ppm
  RMSE     : 22.03 ppm
  Accuracy : 97.4%
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Figure 3: Accuracy results for all three prediction horizons.

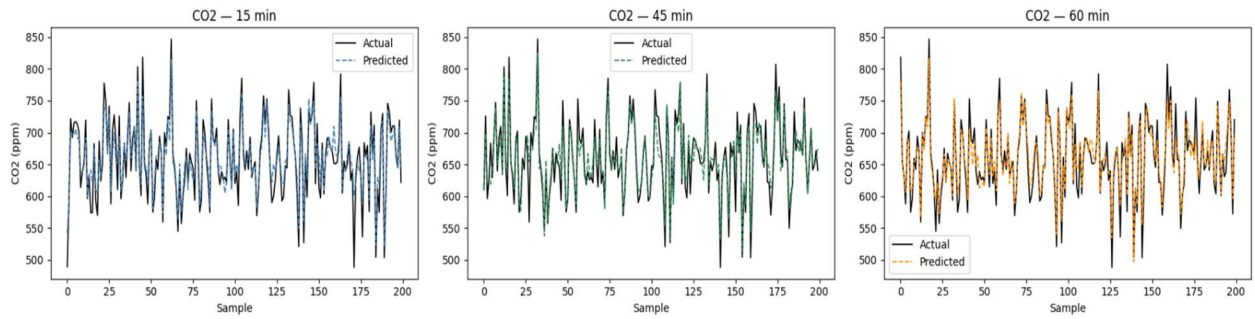

Figure 4: Predicted vs. Actual CO2 values for 15-min, 45-min, and 60-min horizons.

The model achieved a consistent accuracy of 97.4% across all three prediction horizons, with a Mean Absolute Error of approximately 17 ppm. This level of precision is well within the acceptable tolerance for practical indoor air quality management, where intervention thresholds are typically defined at 800 ppm (warning) and 1000 ppm (hazardous).

Notably, the model maintained consistent accuracy across all three prediction horizons, indicating that the learned temporal patterns generalize well beyond the immediate short-term window. This is particularly valuable for proactive ventilation control, where early warnings at the 45-minute and 60-minute horizons allow sufficient time for corrective action.

7. Spike Detection Mechanism

In addition to multi-horizon forecasting, the system incorporates a rule-based CO2 spike detection mechanism that operates independently of the LSTM model. A spike is defined as a sudden increase in CO2 concentration of 200 ppm or more between two consecutive one-minute readings. Such rapid increases may indicate sudden occupancy changes, equipment malfunction, or inadequate ventilation response.

When a spike event is detected, the system immediately triggers a visual alert on the web dashboard. The combination of predictive forecasting and reactive spike detection provides a comprehensive two-layer safety net for indoor air quality management.

8. Online Learning and Room Adaptation

A key feature of the proposed system is its ability to continuously adapt to new environments through online learning. When the sensor is deployed in a new room, the model begins fine-tuning its weights using the incoming real-time data stream, progressively learning the unique CO2 dynamics of that specific space.

Each room adopted model weights are persisted in Firebase Storage, enabling instant model restoration when the sensor is returned to a previously visited room. A weighted learning strategy prioritizes recent data over historical readings, ensuring the model reflects current environmental conditions rather than outdated patterns.

9. Conclusion

This report has presented the design and evaluation of an LSTM-based CO2 prediction system for indoor air quality monitoring. The model achieved 97.4% prediction accuracy across 15-minute, 45-minute, and 60-minute forecasting horizons, with a mean absolute error of approximately 17 ppm. The compact 211 KB model size enables efficient browser-based deployment via TensorFlow.js.

The integration of real-time sensor data acquisition, cloud synchronization, predictive modeling, and anomaly detection within a single unified platform represents a practical and scalable approach to intelligent indoor air quality management. The online learning capability further ensures that the system remains accurate and relevant as environmental conditions evolve over time.

Future work will focus on incorporating additional sensor modalities, extending the prediction horizon, and validating the system across a broader range of indoor environments including classrooms, laboratories, and residential spaces.